THE CATHOLIC UNIVERSITY OF EASTERN AFRICA
Faculty of Science · Department of Physics

Proposal Amendment

Measuring the Invisible: A Sociophysics Framework for Quantifying Emotionally Manipulative Media Content Using Predictive Neural Encoding

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Instrument	`https://monarch.brianmwai.com`
Status	Requires supervisor sign-off before the thesis is written

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1 What this document is

The April proposal was written before the model at the centre of it had been run. Once it was run, three things turned out to be different from what the proposal assumed, and one dataset the proposal names is no longer available to anybody.

None of these is a mistake in the work. Each is a property of the tools, found by testing them. This document states each one in plain terms, says what the project does instead, and asks for four decisions.

The short version: the project still delivers the instrument the proposal promised, and answers three of its four research questions. What it can no longer do is measure the amygdala, because the released model does not predict it.

2 The four changes

2.1 Change 1: the model cannot see the amygdala

What the proposal assumed. That TRIBE v2 predicts activity across 29,286 brain targets including the amygdala and nucleus accumbens. Equation (4) defines the project's central measurement as the ratio of amygdala activity to prefrontal activity.

What the released model actually does. It predicts 20,484 points on the outer surface of the brain only. The amygdala sits deep inside the brain, so it is not among them. This is not a setting that can be switched on; those outputs do not exist in the released weights. It is stated in the checkpoint's own configuration file, which is included with the model.

Why this matters in one sentence. The quantity Equation (4) is built from cannot be computed at all.

What the project does instead. It measures the same balance one level up, on the brain surface: regions associated with emotional salience versus regions associated with deliberate reasoning. Everywhere this appears, it is called a *cortical proxy* and never called the amygdala.

Honest consequence. The thesis can say that emotionally charged content is predicted to shift the balance of surface brain activity. It cannot say anything about the amygdala itself. That limitation is stated in the abstract, not buried.

2.2 Change 2: the original formula does not work on real content

What the proposal specified. A ratio,

$$\mathrm{NAA} \;=\; \frac{A_{\mathrm{amyg}}}{A_{\mathrm{PFC}} + \delta}, \qquad \delta = 10^{-3}.$$

What happens when it is used. The model reports activity in standardised units, centred on zero. That means the bottom of the fraction is negative roughly half the time. Dividing by a negative number flips the answer's sign; dividing by a number near zero makes it explode. On a 40-item test run the formula produced no usable value for a single item.

What the project does instead. It subtracts rather than divides:

$$\mathrm{NAA}_{\mathrm{signed}} \;=\; A_{\mathrm{aff}} - A_{\mathrm{del}}.$$

Positive means the emotional side leads. Negative means the reasoning side leads. This is always defined. The cost is that the number is no longer dimensionless: it carries the units of the model's standardised output, and the thesis says so.

Acknowledgement. This is the change Dr. Mutambi raised at the proposal defence. It is recorded here as her suggestion, adopted once running the model showed why the ratio could not survive standardised output.

2.3 Change 3: the calibration does not generalise

What the proposal specified. Objective (iv): fit the coupling constant $\hat{\alpha}$, the number connecting the brain index to the physics model's field strength.

What happened. It was fitted on two partial runs and then on the completed 400-item corpus. On the partial runs the confidence interval comfortably included zero. On the full corpus it no longer does. That sounds like a result and is not one: the fit still performs *worse than the sample mean* on held-out data, so it has no out-of-sample predictive value, and the estimate is not identified on its own. It scales inversely with the coupling βJ, which no measurement in this study fixes, so any single value states a chosen βJ rather than anything about media.

What the project does instead. It quotes no value for $\hat{\alpha}$ anywhere – not in a table, not in a caption, not "for illustration". The phase structure is then presented across a range of possible α values, which shows how the physics would behave for each, and states that the data do not pin it down.

Why this is acceptable physics. A properly powered measurement constrains what everyone working after you has to believe, whichever way it comes out. The sample size and the smallest detectable effect are both reported, so a reader can see what the design could and could not have found. Reporting an unidentified coupling as though it were a measured one would be the failure here, not declining to quote it.

2.4 Change 4: a named dataset was withdrawn by its authors

What the proposal specified. NELA-GT-2021, for the credibility scores in objective (iv).

What happened. Its authors removed it from public access on 1 January 2024, after the proposal was written. It now requires per-application manual approval. This was verified rather than assumed: with a valid Harvard Dataverse account, a control file from an unrelated dataset downloads normally while every NELA file returns "forbidden", and the dataset does not appear in search at all.

What the project does instead. It uses SemEval-2019 Task 4 (Hyperpartisan News Detection), openly licensed CC BY 4.0. This is arguably a better fit: NELA labels whether a *publisher* is credible, while this labels the *article*, and the project's categories are defined by what an article does.

3 What the thesis will deliver

3.1 Delivered

1. **The measurement instrument, working and public.** The proposal calls this the primary deliverable and says so three times. It exists: the batch pipeline, the four-category corpus, and the corpus-level audit report described in §5.8(iii) – ranked table of every item, distribution per category, free-energy landscape, and per-item flags. Released as open-source code with the corpus manifest and one complete example report. Live at `https://monarch.brianmwai.com`.
2. **The physics, derived in full.** The mean-field Ising treatment, the self-consistency condition, the free-energy expansion, the susceptibility and the phase structure. This stands on its own and does not depend on any experimental outcome. One coefficient error in the proposal's Equation (10) is corrected in the derivation.
3. **A measured result on 400 items.** One hundred items in each of four categories, length-matched, with wire-service tells removed and sources balanced within each category, so that the categories differ by content rather than by where they came from.
4. **Two findings that are new to the literature.** First, that the released TRIBE v2 checkpoint predicts only surface activity, which constrains any study built on it. Second, that ratio-type measures over standardised model output are undefined on real content. This is the first independent research application of this model.

3.2 Not delivered, and why

1. **Any statement about the amygdala.** The instrument cannot reach it.
2. **A calibrated value for $\hat{\alpha}$.** The fit is not distinguishable from zero, so quoting a number would misrepresent it.
3. **1,500 items across text, audio and video.** The corpus is 400 text items. Each item takes about 75 seconds of GPU time, and the available quota does not cover the original size. Text is the modality the proposal processes first by design.
4. **Research Question IV.** It compares information across modalities, which requires the audio and video that were dropped.

3.3 Status of each objective

#	Objective	Status after this amendment
(i)	Build the corpus	Delivered at 400 text items, not 1,500 multi-modal
(ii)	Run TRIBE, compute the index	Delivered, using the signed index
(iii)	Characterise the distribution	Delivered
(iv)	Calibrate $\hat{\alpha}$	Attempted; reported as unidentified, no value quoted
(v)	Derive the phase structure	Delivered across a range of α, not at one fitted value
(vi)	Validate as a classifier	Delivered, reporting AUC as the headline figure
(vii)	Deliver the instrument	Delivered; this is the primary deliverable

3.4 What an examiner receives

A thesis that sets out a physics framework, builds the instrument the proposal promised, runs it on a controlled corpus, and reports honestly what it found and what it could not reach. On the completed 400-item corpus the index *does* separate the four categories, at an effect size the design was powered to detect. The separation is carried almost entirely by one category, and it cannot be attributed to framing: each category is drawn from a single source dataset, so category and provenance are confounded by construction. The finding is reported with that confound stated first, alongside the sample size and the smallest effect the design could have detected.

4 Decisions requested

1. Approve the amended measurement (cortical proxy, signed difference) and the corresponding rewrite of the abstract, §4.1 and Equation (4). *Blocking.*

2. Approve reducing the corpus to 400 text items and dropping audio and video, with Research Question IV declared out of scope. *Blocking.*

3. Confirm that the coupling is reported as unidentified, with no value of $\hat{\alpha}$ quoted anywhere. *Blocking.*

4. Note the substitution of SemEval-2019 Task 4 for the withdrawn NELA-GT-2021.

Brian Mwai (1050555) Date Dr. Songa Mutambi Date